

BMJ Open Diagnostic yield and safety of probe-based confocal laser endomicroscopy-guided pleural biopsy via semirigid thoracoscope in diagnosing patients with unknown pleural effusion: a protocol for a multicentre, randomised controlled trial (COLLABORATION-I)

Ming-Ming Deng,¹ Yi Hu,² Xiaoping Liu,³ Yu Feng,⁴ Ziwen Zheng ,^{1,5} Run Tong,¹ Jose Porcel,⁶ Gang Hou ¹

To cite: Deng M-M, Hu Y, Liu X, *et al*. Diagnostic yield and safety of probe-based confocal laser endomicroscopy-guided pleural biopsy via semirigid thoracoscope in diagnosing patients with unknown pleural effusion: a protocol for a multicentre, randomised controlled trial (COLLABORATION-I). *BMJ Open* 2026;**16**:e117208. doi:10.1136/bmjopen-2026-117208

► Prepublication history and additional supplemental material for this paper are available online. To view these files, please visit the journal online (<https://doi.org/10.1136/bmjopen-2026-117208>).

M-MD, YH, XL, YF and ZZ contributed equally.

Received 21 January 2026
Accepted 22 May 2026



© Author(s) (or their employer(s)) 2026. Re-use permitted under CC BY-NC. No commercial re-use. See rights and permissions. Published by BMJ Group.

For numbered affiliations see end of article.

Correspondence to

Dr Gang Hou;
hougangcmu@163.com

ABSTRACT

Introduction Semirigid thoracoscopy plays an important role in the diagnosis of pleural diseases. However, its diagnostic performance remains unsatisfactory particularly in terms of the negative likelihood ratio. Therefore, more effective supplementary diagnostic tools are required. Probe-based confocal laser endomicroscopy (pCLE), which allows live tissue imaging at the cellular level, can discriminate between malignant and benign pleura during medical thoracoscopy. However, the clinical relevance of pCLE in pleural disease remains unclear. This protocol describes a randomised controlled trial that evaluates the additional diagnostic value of pCLE in diagnosing pleural diseases using semirigid thoracoscopy.

Methods and analysis This study is a multicentre, parallel-group, randomised controlled trial that will be conducted at ten sites in China. A total of 158 adult patients with undiagnosed exudative pleural effusions will be enrolled and randomly allocated (1:1) to undergo either a conventional pleural biopsy (control group) or a pCLE-guided pleural biopsy (intervention group) via semirigid thoracoscopy. In the intervention group, a pCLE system will be applied during thoracoscopy to identify suspicious pleural areas for targeted biopsy. The primary outcome is the diagnostic yield of the procedure in patients with unknown causes of pleural effusion. Secondary outcomes include negative likelihood ratio, diagnostic sensitivity in specific diseases, procedural time, rate of adequate specimens for achieving molecular diagnosis and complications.

Ethics and dissemination Ethics approval was obtained from the China-Japan Friendship Hospital Ethics Committee (2025-KY-018). Written informed consent will be obtained from all the participants. The findings will be disseminated through journal publications and conference presentations.

Trial registration number NCT06741839.

STRENGTHS AND LIMITATIONS OF THIS STUDY

- ⇒ To ensure consistent intervention quality and minimise operator-dependent bias, all participating physicians are required to complete a standardised training programme and pass a competency assessment in probe-based confocal laser endomicroscopy (pCLE) image interpretation.
- ⇒ Implementing a 12-month follow-up for participants initially diagnosed with non-specific or benign histology improves the reliability of the final reference diagnosis, thereby enhancing the validity of the primary outcome assessment.
- ⇒ The methodological design is limited by the absence of stratification during randomisation based on pleural morphology, as well as the inherent impossibility of blinding the proceduralists to the biopsy technique, which potentially introduces performance bias.
- ⇒ The added diagnostic benefit of pCLE is minimal for patients with visible tumours under white light thoracoscopy, but it is potentially valuable for those with normal or slightly abnormal pleura. The trial lacks the power for subgroup analysis based on thoracoscopic appearance, making any such findings exploratory and in need of future validation.

INTRODUCTION

Medical thoracoscopy is widely recommended as the technique of choice for diagnosing exudative pleural effusions of uncertain aetiology.^{1 2} Current medical thoracoscopy can be performed using rigid or semirigid thoroscopes.³ While semirigid thoracoscopy offers advantages such as enhanced flexibility and patient tolerance, several studies have suggested that the diagnostic yield remains

suboptimal in the intention-to-treat population.^{4 5} Moreover, the negative predictive value of semirigid thoracoscopy is low, and a 15.4% rate of delayed malignancy diagnosis has been reported among patients with initially benign histology results (nonspecific pleuritis/fibrosis) via semirigid thoracoscopy.^{6–8} This may be associated with the biopsy success rate, sampling quality and visual findings during the thoracoscopy.

For patients with chronic pleuritis or pleuritis due to asbestos exposure, obtaining biopsy specimens using semirigid instruments and flexible forceps is more difficult because of the lack of mechanical power.^{9 10} To overcome the disadvantages of semirigid thoracoscopy and improve the diagnostic yield, previous studies have introduced cryobiopsy and electrocautery-assisted biopsy to obtain larger specimens.^{11 12} Despite a recent meta-analysis showing a significant improvement in sampling quality, no difference was observed in diagnostic yield.¹³ Thoracoscopic findings also significantly affect the diagnostic yield of semirigid thoracoscopy. A previous study suggested that the presence of nodules predicts a high probability of obtaining a definitive histological diagnosis on pleural biopsy specimens, while it remains difficult to select the sampling location in patients with diffuse pleural thickening.⁴ These findings suggest that improving the accessibility of representative specimens rather than sampling quality is important for improving the diagnostic value of semirigid thoracoscopy. Although novel imaging techniques, such as autofluorescence imaging (AFI), can improve the negative predictive value, their specificity is limited by the interference of tissue component heterogeneity (eg, fibrous plaques and necrotic tissue).^{14 15}

Probe-based confocal laser endomicroscopy (pCLE) is a real-time cellular imaging technique that can achieve 'optical biopsy' in vivo. Bonhomme *et al* demonstrated distinct features between benign and malignant pleural involvement and showed the potential of pCLE as a guidance tool for pleural biopsy.¹⁶ In addition, a prospective study assessed the use of pCLE to guide biopsies in the specific indication of suspected pleural mesothelioma and suggested that pCLE could distinguish malignant mesothelioma from pleural fibrosis with high precision.¹⁷ Therefore, we hypothesised that pCLE could improve the diagnostic yield by sampling representative areas detected by pCLE imaging of the pleura. However, evidence on the diagnostic yield and safety of pCLE-guided pleural biopsy using a semirigid thoracoscope with flexible forceps is lacking.

Hence, the COLLABORATION-I trial aims to evaluate whether pCLE-guided biopsy via semirigid thoracoscopy can improve diagnostic yield while maintaining safety.

METHODS AND ANALYSIS

Study design

This inferiority randomised parallel study will be conducted in eight experienced diagnosis and treatment

centres for pleural disease teaching hospitals in China (China-Japan Friendship Hospital, Shanghai Ruijin Hospital, Henan Provincial People's Hospital, Second Affiliated Hospital of Harbin Medical University, Anhui Chest Hospital, Wuhan Pulmonary Hospital, The Second People's Hospital of Gansu Province, Shandong Public Health Clinical Centre). All participating sites are equipped with a semirigid thoracoscope and pCLE probe.

Enrolment began on 11 March 2025, with the first patient recruited that day. The trial aims to finish by May 2027 and is actively recruiting participants.

Selection criteria

Inclusion criteria

Patients included in this trial will meet the following criteria:

1. Age ≥ 18 years.
2. Evidence of exudative pleural effusion in which a specific diagnosis could not be determined using clinical, radiological, laboratory or cytological examinations.
3. Willingness to participate in the study and undergo an invasive procedure.

Exclusion criteria

Patients who meet any of the following criteria will not be eligible for enrolment in this trial:

1. Pleural thickening or pleural-based mass without pleural effusion on radiologic examination.
2. Presence of parapneumonic effusion.
3. Any contraindication to pleural biopsy or semirigid thoracoscopy.
4. Participation in other studies within 3 months without withdrawal or termination.

Sample size calculation

This is a non-inferiority study with a 10% margin. It should be considered that the diagnostic yield of semirigid thoracoscopy with flexible forceps in patients with exudative pleural effusion is approximately 93%, and our initial experience showed a diagnostic yield of 95% for pCLE-guided pleural biopsy via semirigid thoracoscopy.¹⁸ Assuming a dropout rate of 20%, a total of 158 patients with exudative pleural effusion need to be included in this study to achieve a precision level of 0.9 with $\alpha=0.05$. PASS was used to calculate the sample size (figure 1).

Randomisation and grouping

Eligible patients will be randomly assigned to the conventional semirigid thoracoscopy and pCLE-guided semirigid thoracoscopy groups at a 1:1 ratio. An independent investigator from the chief centre (China-Japan Friendship Hospital) will use a Sealed Envelope online randomisation programme with block sizes of four (<https://www.sealedenvelope.com/simple-randomiser/v1/lists>) to generate the randomisation sequence. The randomisation sequence and block size are concealed from the study investigators and staff.

Non-Inferiority Tests for the Difference Between Two Proportions

Numeric Results

Test Statistic: T-Test

Hypotheses: H0: $P_1 - P_2 \leq D_0$ vs. H1: $P_1 - P_2 > D_0$

Target Power	Actual Power*	N1	N2	N	Ref. P2	P1 H0 P1.0	P1 H1 P1.1	NI Diff D0	Diff D1	Alpha
0.9	0.90281	66	66	132	0.927	0.827	0.95	-0.1	0.023	0.05

Figure 1 Results of sample size calculation based on the non-inferior design for comparison of diagnostic yield in the whole population.^{36–44} *Power was computed using the normal approximation method.

Recruitment process

Eligible patients will be identified by attending physicians from both outpatient clinics and inpatient wards across the eight participating centres. Subsequently, the study team will screen these patients against the established eligibility criteria, furnish them with comprehensive study information and secure written informed consent before initiating any study-related procedures. Recruitment will be conducted on an ongoing basis until the target sample size of 158 patients is achieved.

Intervention procedure

(1) A total of 158 patients with exudative pleural effusion who meet the inclusion criteria will be enrolled in the study after providing written informed consent. No financial or other incentives will be offered to participants for their involvement in the study. Reimbursement of reasonable travel expenses will be provided when applicable.

(2) All participants will undergo complete blood count, liver and renal function tests, coagulation profile determination and electrocardiography. The size of effusions on chest radiographs is commonly graded, both in daily practice and research studies, according to a semiquantitative rule described in a previous study,¹⁹ as follows: 0 (no effusion), 1 (blunting of the costophrenic angle), 2 (<25% of the hemithorax but more than blunting), 3 (25%–50% of the hemithorax), 4 (50%–75% of the hemithorax) and 5 (>75% of the hemithorax).

(3) After excluding contraindications, participants will undergo semirigid thoracoscopy under mild sedation and local anaesthesia. Randomisation envelopes will be opened immediately prior to the procedure, and all participants will receive the randomly assigned procedure (conventional or pCLE-guided pleural biopsy via semirigid thoracoscopy) according to the protocol.

(4) Procedure: patients with spontaneous breathing (fasting for 8 hours) under conscious sedation using incremental doses of midazolam 0.5–5 mg and observing complete aseptic precautions in a bronchoscopy suite will undergo thoracoscopy. Supplemental oxygen will be administered using an air-entrainment mask. Blood pressure will be monitored every 10 min, and oxygen saturation, heart rate and rhythm will be continuously monitored. The operators are consultant pulmonologists or pulmonary fellows with 2.5 years of training and assisted in at least 25 procedures under the direct supervision of one of the consultants. A standard training course in pCLE image identification will be used to train the

thoracoscopists; only thoracoscopists who pass the final examination will be allowed to participate in this trial.

The patient is first placed in the lateral decubitus position with the affected side up and the arm raised above the head. A single-entry site is selected in the fourth or fifth intercostal space along the midaxillary line. The site is infiltrated with 2% lidocaine, and a 1–3 cm skin incision is made. Blunt dissection will be used to enter the pleural space, and the incision will be extended if necessary. A flexible trocar will be inserted, followed by the introduction of a semirigid thoracoscope (LTF-260; Olympus, Tokyo, Japan). The pleural fluid is aspirated, and air is allowed to enter the pleural space.

For patients assigned to receive pleural biopsy via semirigid thoracoscopy alone, the parietal, visceral and diaphragmatic pleura will be inspected, and eight biopsies will be obtained under direct vision in all suspected areas of the parietal pleura.

For patients assigned to receive pCLE-guided pleural biopsy via semirigid thoracoscopy, pCLE will be introduced to detect suspected areas of the parietal pleura. The pCLE inspection will follow a standardised protocol with two main steps. First, any visible abnormalities under white light, such as nodules or thickening, will be examined with pCLE to evaluate intralesional heterogeneity and determine the best biopsy site. Second, the basal pleura will be systematically swept, covering at least two intercostal spaces above the costophrenic recess, including the diaphragmatic pleura and lower posterior chest wall, which are common sites for malignant involvement.^{20 21} Real-time pCLE images will be analysed for abnormal cellular patterns, guiding biopsy to these areas.¹⁶ The pCLE images will be interpreted in real-time by the operator during the procedure. Abnormal findings in pCLE are delineated according to established criteria from prior studies. Indicators of malignancy encompass pleomorphic cells exhibiting variability in size and shape, indistinct intercellular borders, a clustered or papillary architectural pattern and the disruption of the normal mesothelial arrangement. Conversely, features indicative of tuberculous or necrotic lesions include unstructured regions, disrupted cellular architecture and a heterogeneous inflammatory milieu.^{17 22} The identification of any such abnormal characteristics designates the respective pleural region as the target for biopsy. After confirming the targeted area of the parietal pleura for pleural biopsy, the pCLE probe will be withdrawn, and a flexible forceps

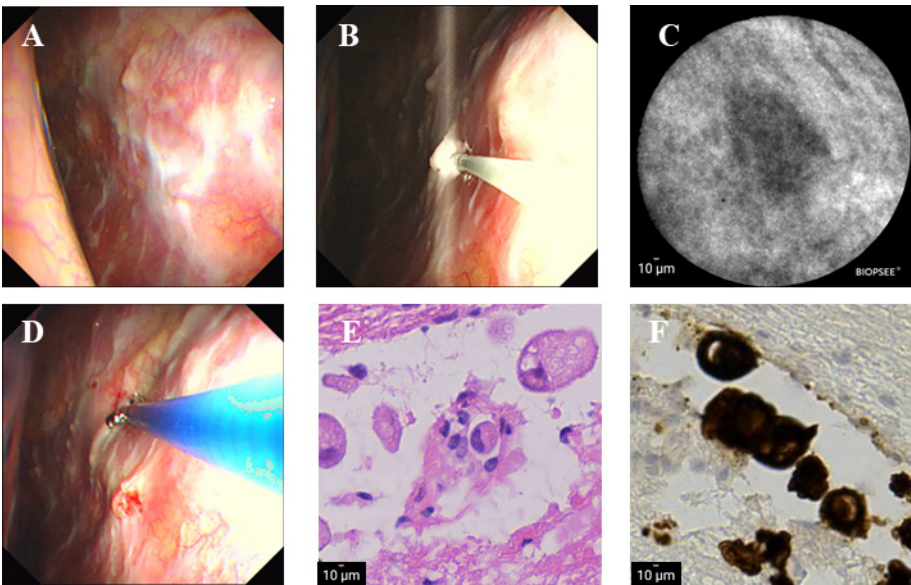


Figure 2 Schematic flow of pCLE-guided pleural biopsy via semirigid thoracoscopy. (A) Multiple small pleural nodules with fibrotic formation detected using semirigid thoracoscopy; (B) pCLE probe detecting pleural nodules via semirigid thoracoscopy; (C) detection of tumour deposits in the pleura by in vivo pCLE imaging, visualising tumour cells; (D) obtaining tissue using flexible forceps in the same location where pCLE detected the specific images; (E) a representative H&E staining image showing the existence of tumour cells; (F) these tumour cells showed a positive reaction for CK7. pCLE, probe-based confocal laser endomicroscopy.

biopsy will be performed (figure 2A–F). This process will be repeated eight times, and eight samples will be obtained. All specimens will be transported in separate formalin containers. with biopsy sequences for histopathological analysis.

If an adequate pleural space could not be created, the procedure will be stopped. A 16–20French intercostal drainage tube will be inserted before wound closure to evacuate air and fluid. If the thoracoscopic findings suggest malignancy (variable-sized nodules), iodopovidone (20mL of 10% solution) pleurodesis will be performed during the procedure or later through a chest tube. Chest radiographs will be obtained 3 hours after the procedure. The chest tube will be removed only once the drainage decreases to <50 mL/d for two consecutive days, provided the effusion is non-purulent.

(5) Morphometrical and morphologic analysis of thoracoscopic biopsies: all biopsies will be processed conventionally by serial sectioning of at least eight H&E-stained sections, as per the standard procedure. All biopsies will be surveyed in terms of size and quality. The pathologist will be blinded to the technique used for obtaining the biopsies. For each patient, a definitive diagnosis will be made based on the pathology results. In addition, molecular analysis of malignant pleural diseases will be conducted as part of routine practice. The definitions of suitable tumour types and related molecular markers were consistent with those of a previous study⁸ (table 1).

(6) Follow-up: a follow-up period of 12 months will be performed for every patient who receives a benign pleural effusion diagnosis without direct evidence for a definitive diagnosis or who is still suspected of having a malignant

pleural effusion, despite the absence of malignant histopathological results in their biopsy tissue. Patients who remain undiagnosed with a specific benign or malignant disease after 12 months will be classified as having ‘idiopathic cases’.

The trial flowchart is shown in figure 3.

Reference standard

The diagnosis in this trial will be established by the treating clinician based on all clinical, radiological, microbiological and pathological information. In addition, all the final diagnoses will be verified by an independent

Tumour type	Recommended predictive marker
NSCLC	PD-L1, EGFR, ALK, ROS-1
Breast	HER-2, ER, PR
Ovarian	BRCA
Renal	PD-L1
DLBCL	MYC, BCL2, BCL6
Colorectal	RAS, BRAF, EGFR
Gastric	HER2
Oesophageal	HER2
Melanoma	BRAF, NRAS, cKit, PD-L1

ALK, anaplastic lymphoma kinase; DLBCL, diffuse large B-cell lymphoma; EGFR, epidermal growth factor receptor; ER, oestrogen receptor; HER2, human epidermal growth factor receptor 2; NSCLC, non-small cell lung cancer; PD-L1, programmed death ligand 1; PR, progesterone receptor; ROS-1, C-ros oncogene.

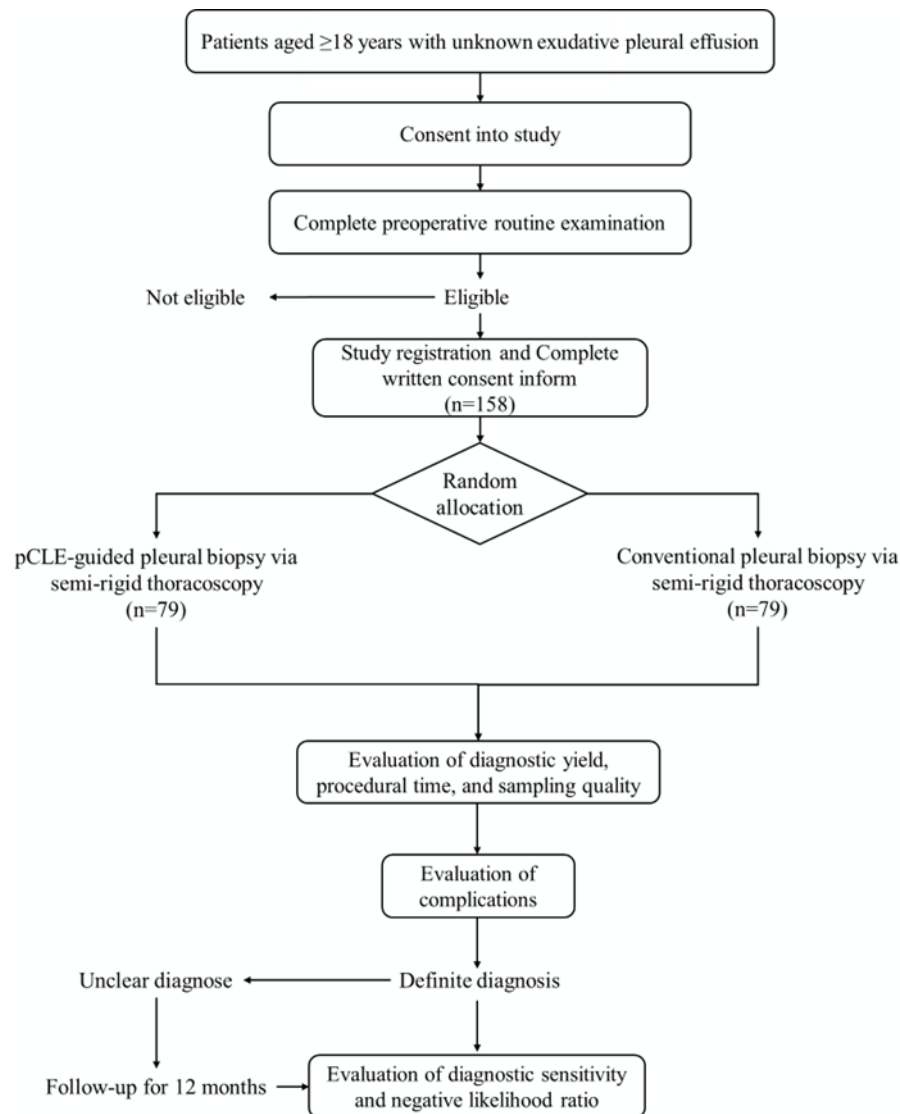


Figure 3 Trial flow chart. pCLE, probe-based confocal laser endomicroscopy.

diagnosis verification panel, which includes three pulmonary and critical care medicine physicians, one radiologist and one pathologist.

Outcome

Primary outcome

The primary outcome is the diagnostic yield of conventional pleural biopsy via semirigid thoracoscopy and pCLE-guided pleural biopsy via semirigid thoracoscopy in patients with unknown exudative pleural effusion.²³ The diagnostic yield in this study is identified as the proportion of all individuals undergoing pleural biopsy under evaluation in whom a specific diagnosis is established.²⁴ Based on the existing literature and our preliminary experience, the expected diagnostic yield is approximately 93% for the control group (conventional pleural biopsy via semirigid thoracoscopy) and 95% for the intervention group (pCLE-guided pleural biopsy via semirigid thoracoscopy). A pre-established non-inferiority margin of 10% has been set for this comparison.

Secondary outcome

The secondary outcomes are diagnostic sensitivity for specific diseases (including pleural mesothelioma, metastatic cancer and tuberculous pleurisy), procedure time, rate of adequate specimens for achieving molecular diagnosis and incidence of any complications.

Complications related to the procedure include prolonged air leaks, haemorrhage, subcutaneous emphysema, postoperative fever, empyema, wound infection, cardiac arrhythmias, hypotension and seeding of the chest wall from mesothelioma. The biopsy sites will be evaluated for haemorrhage prior to performing subsequent biopsies, and the degree of bleeding at the biopsy site will be assessed as follows²⁵:

1. Nil: slight and self-limiting.
2. Mild bleeding requiring local injection of vasoactive medication (epinephrine).
3. Moderate to severe bleeding: requiring electrocautery or argon plasma coagulation.

Assessment of sampling quality: The biopsy quality will be determined based on the tissue depth. A thoracoscopic biopsy specimen, including fatty tissue from the thoracic wall, is considered a deep biopsy and is therefore of high quality. The tissue depth of each specimen will be recorded.

The interpretability of the specimens will be assessed as follows.

1. Easily interpretable: sufficient tissue with all elements required for diagnosis.
2. Interpretability with some difficulty: less tissue or fewer diagnostic elements and less reliable diagnosis.
3. Interpretation with great difficulty: little tissue or scant diagnostic elements and low reliability of diagnosis.
4. Non-interpretable: diagnosis was not possible.

The definitions for the molecular marker status are agreed on in a previous study.²⁶ The following criteria were used to classify each diagnostic biopsy into one of the three categories:

1. Successful molecular marker analysis: if the pathologist reports a successful molecular marker analysis of the biopsy specimen. If the pathologist does not run all of the molecular tests for the specific tumour type, as recommended by Tsim *et al*²⁷ provided the report does not explicitly describe a failure in any of the tests performed, the result will be considered successful.
2. Unsuccessful molecular marker analysis: if the pathologist explicitly reports an inability to conduct a molecular marker test (ie, test failure).
3. Not sent, despite relevance: if the pathologist does not perform molecular analysis despite it being of relevance to the particular tumour type, as recommended by Tsim *et al*.^{28–30}

Statistical analysis

According to the Reporting Standards for Diagnostic Testing, patients diagnosed with nonspecific pleurisy after 12 months of follow-up will be excluded from the calculation of diagnostic yield. Sensitivity is defined as the proportion of individuals with a specific disease (malignant pleural mesothelioma, metastatic cancer or tuberculous pleurisy), as determined by the reference standard, who have a positive test result. The negative likelihood ratio will be calculated as $(1 - \text{sensitivity}) / (\text{specificity})$. Because there will be no false-positive findings in the biopsy results, this value will equal $(1 - \text{sensitivity})$. In addition, a higher-dimensional contingency table will be used in this study to demonstrate the results of each biopsy method.²⁴ The accuracy value with a 95% CI will be reported in this study. Subgroup analysis will be conducted according to pleural thickness and lesion morphology. In addition, an exploratory subgroup analysis will be conducted based on the white light semirigid thoracoscopic appearance, categorising cases into those with obvious tumour deposits visible and those with normal or slightly abnormal pleura. Then, the difference in diagnostic accuracy between pCLE-guided pleural biopsy via semirigid

thoracoscopy and conventional pleural biopsy via semirigid thoracoscopy will be evaluated.

The chi-square test will be used to analyse the relationships between the categorical variables. For continuous variables, differences between three or more groups will be assessed using one-way analysis of variance and post-hoc Tukey's multiple comparison test (for normally distributed data) or the Kruskal-Wallis test (for non-normally distributed data). Differences between the two groups will be assessed using the t-test (for normally distributed data) or Mann-Whitney test (for non-normally distributed data). Statistical significance will be set at $p < 0.05$. Statistical analyses will be performed using SPSS software (V.13.0; IBM). The final results of this trial will be reported following the STARD checklist. The STARD reporting guideline is available at <https://www.equator-network.org/reporting-guidelines/stard/>.

Data collection and management

Each site will use standardised case report forms to gather baseline and outcome data. Outcome assessors will receive training, and pathologists will not know the group assignments. Data will be input into a centralised electronic database that is password-protected, with double entry and range checks, anonymised and stored securely at the coordinating centre. Data collection forms can be obtained from the corresponding author.

Data monitoring

A formal data monitoring committee will not be set up for this trial due to the established safety of semirigid thoracoscopy and pCLE, which are minimally invasive and have a low risk of serious adverse events. Instead, an independent endpoint adjudication committee will periodically review safety data and report serious adverse events to the ethics and steering committees. The steering committee will make safety-related decisions based on these reviews. No interim analyses or early stopping rules are planned. Final analyses will occur after the last patient completes follow-up. Adverse events will be recorded through both solicited and spontaneous reports, with serious events reported to the coordinating centre.

ETHICS AND DISSEMINATION

This study has obtained the ethics approval from China-Japan Friendship Hospital Ethics Committee (2025-KY-018). Written informed consent will be obtained from all patients or their guardians before enrolment in the study. A sample participant consent form is provided as online supplemental material. This study will be conducted according to the principles of the Declaration of Helsinki and the International Council for Harmonisation Guidelines for Good Clinical Practice. If the protocol is revised, this information will be promptly shared with the co-researchers. The results of

this trial will be submitted to a peer-reviewed journal for publication. The results of this trial will be disseminated through peer-reviewed publications and conference presentations.

Patient and public involvement

The patients and the public are not involved in the design, conduct, reporting or dissemination of our research plans.

DISCUSSION

This multicentre randomised controlled trial was designed to evaluate whether pCLE can improve the diagnostic accuracy (including diagnostic yield, sensitivity and negative likelihood ratio) of pleural biopsies performed using semirigid thoracoscopy in patients with undiagnosed exudative pleural effusion. While semirigid thoracoscopy is a well-established, minimally invasive technique with acceptable safety, challenges in diagnostic accuracy remain. This limitation is largely attributed to sampling errors and the mechanical constraints of flexible forceps compared with rigid instruments.

Several alternatives have been reported in previous studies, such as cryobiopsy and electrocautery-assisted biopsies. These methods could increase specimen size and quality but have not consistently translated into higher diagnostic yields.^{31 32} Similarly, adjunctive imaging modalities,^{33 34} such as AFI and narrow-band imaging, have shown potential to enhance the negative predictive value of thoracoscopy. However, their specificity is often compromised by confounding tissue characteristics such as fibrosis, necrosis and haemorrhage. Rapid on-site evaluation may offer additional benefits in selected cases but is limited by interpretative challenges, particularly in distinguishing reactive mesothelial proliferations from malignancy and in the context of ongoing pleural repair.³⁵

pCLE offers a distinct advantage in this context by providing real-time in vivo microscopic imaging, which enables more precise targeting of biopsy sites based on cellular features rather than gross morphological cues. Previous studies have demonstrated its high sensitivity and specificity in differentiating malignant from benign pleural involvement, and our preliminary observations suggest that it may detect malignant cells even within fibrotic tissue. Consequently, the primary inquiry is whether pCLE provides additional diagnostic value beyond that of white light semirigid thoracoscopy. The application of pCLE is limited, as biopsies can be directly obtained from these observable lesions. Conversely, in patients exhibiting normal or only slightly abnormal parietal pleura, which is often characterised by non-adhesive fibrotic changes or subtle thickening, malignant or granulomatous cells may still be present at the cellular level despite an unremarkable macroscopic appearance. In such conditions, random

or visually unguided biopsies are prone to missing the target, resulting in a high false-negative rate. Therefore, we posit that pCLE will be most beneficial in the subgroup of patients lacking obvious tumour deposits, particularly when fibrosis obscures the lesion boundaries. However, existing evidence is limited to small, single-centre studies.¹⁷ To evaluate this hypothesis, a pre-specified exploratory subgroup analysis based on thoroscopic white light appearance, differentiating between the presence and absence of specific pathology findings

This trial incorporates several methodological safeguards to optimise internal validity. Participating physicians will undergo standardised pCLE training and competency assessments to reduce operator-dependent variability. Subgroup analyses will be conducted according to pleural thickness, with or without pleural nodularity, to explore whether pCLE is particularly advantageous for malignant pleural effusions with fibrotic adhesions that are difficult to localise visually.

This protocol has several limitations. First, physicians without pCLE experience may find it difficult to identify pCLE features during the procedure. To ensure that biopsies are taken from areas with abnormal pCLE images, standardised training and a pCLE image recognition test will be conducted, and only those who pass will participate. Second, stratified randomisation based on thoroscopic appearance is challenging in practice and will not be used in this study design. We hypothesised that pCLE primarily improves the diagnostic efficiency of semirigid thoracoscopy by detecting invisible lesions. However, differences in preoperative preparation (sodium fluorescein injection in the pCLE group), lack of correlation between chest CT and thoroscopic appearance, and the absence of a standard classification system make stratification impractical. Instead, subgroup analyses will be performed based on pleural thickness and lesion type (nodular/diffuse) to assess whether pCLE benefits the diagnosis of small pleural lesions. Additionally, an exploratory subgroup analysis will be conducted based on the semirigid thoroscopic appearance under white light, distinguishing between cases with evident tumour deposits and those with normal or slightly abnormal pleura. It is important to acknowledge that this trial is not statistically powered to support this comparison, and therefore, the results should be interpreted as exploratory.

Author affiliations

¹National Center for Respiratory Medicine; State Key Laboratory of Respiratory Health and Multimorbidity; National Clinical Research Center for Respiratory Diseases; Institute of Respiratory Medicine, Chinese Academy of Medical Sciences; Department of Pulmonary and Critical Care Medicine, Center of Respiratory Medicine, China-Japan Friendship Hospital, Beijing, People's Republic of China

²Department of Pulmonary and Critical Care Medicine, The Central Hospital of Wuhan, Wuhan, People's Republic of China

³Department of Pulmonary and Critical Care Medicine, Second People's Hospital of Gansu Province, Lanzhou, People's Republic of China

⁴School of Biomedical Engineering, Tsinghua University, Beijing, People's Republic of China

⁵China-Japan Friendship Hospital (Institute of Clinical Medical Sciences), Chinese Academy of Medical Sciences, Beijing, People's Republic of China

⁶Pleural Medicine and Clinical Ultrasound Unit, Department of Internal Medicine, Arnau de Vilanova University Hospital, IRBLleida, University of Lleida, Lleida, Spain

Contributors GH, MM-D, YH, XL and YF conceived the protocol design for this randomised controlled trial, and GH is the guarantor. GH, MM-D, YH, XL and YF drafted the manuscript and revised it. ZZ and RT are responsible for data collection and preparation. JP critically revised the trial design. All authors reviewed several drafts of the manuscript and approved the final version.

Funding This work is financially supported by the National High-Level Hospital Clinical Research Funding (No. 2025-NHLHCRF-JBGS-A-WZ-14), Excellence & Innovation Initiative of China-Japan Friendship Hospital (No. ZRZC2025-XYA01) and Noncommunicable Chronic Diseases-National Science and Technology Major Project (2024ZD0522800 to P Tian, 2024ZD0522804 to M Deng). The funders have no role in the study design, data collection, management, analysis, interpretation, report writing or the decision to submit the findings for publication, and will not have ultimate authority over any of these activities.

Competing interests None declared.

Patient and public involvement Patients and/or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research.

Patient consent for publication Not applicable.

Provenance and peer review Not commissioned; externally peer reviewed.

Supplemental material This content has been supplied by the author(s). It has not been vetted by BMJ Publishing Group Limited (BMJ) and may not have been peer-reviewed. Any opinions or recommendations discussed are solely those of the author(s) and are not endorsed by BMJ. BMJ disclaims all liability and responsibility arising from any reliance placed on the content. Where the content includes any translated material, BMJ does not warrant the accuracy and reliability of the translations (including but not limited to local regulations, clinical guidelines, terminology, drug names and drug dosages), and is not responsible for any error and/or omissions arising from translation and adaptation or otherwise.

Open access This is an open access article distributed in accordance with the Creative Commons Attribution Non Commercial (CC BY-NC 4.0) license, which permits others to distribute, remix, adapt, build upon this work non-commercially, and license their derivative works on different terms, provided the original work is properly cited, appropriate credit is given, any changes made indicated, and the use is non-commercial. See: <https://creativecommons.org/licenses/by-nc/4.0/>.

ORCID iDs

Ziwen Zheng <https://orcid.org/0000-0003-1711-6328>

Gang Hou <https://orcid.org/0000-0003-3438-1764>

REFERENCES

- Roberts ME, Rahman NM, Maskell NA, *et al*. British Thoracic Society Guideline for pleural disease. *Thorax* 2023;78:s1–42.
- Scherpereel A, Opitz I, Berghmans T, *et al*. ERS/ESTS/EACTS/ESTRO guidelines for the management of malignant pleural mesothelioma. *Eur Respir J* 2020;55:1900953.
- Rahman NM, Ali NJ, Brown G, *et al*. Local anaesthetic thoracoscopy: British Thoracic Society Pleural Disease Guideline 2010. *Thorax* 2010;65 Suppl 2:ii54–60.
- Dhooria S, Singh N, Aggarwal AN, *et al*. A randomized trial comparing the diagnostic yield of rigid and semirigid thoracoscopy in undiagnosed pleural effusions. *Respir Care* 2014;59:756–64.
- Dhooria S, Bal A, Sehgal IS, *et al*. Pleural Cryobiopsy versus Flexible Forceps Biopsy in Subjects with Undiagnosed Exudative Pleural Effusions Undergoing Semirigid Thoracoscopy: A Crossover Randomized Trial (COFFEE Trial). *Respiration* 2019;98:133–41.
- Heilo A, Stenwig AE, Solheim OP. Malignant pleural mesothelioma: US-guided histologic core-needle biopsy. *Radiology* 1999;211:657–9.
- Davies HE, Nicholson JE, Rahman NM, *et al*. Outcome of patients with nonspecific pleuritis/fibrosis on thoracoscopic pleural biopsies. *Eur J Cardiothorac Surg* 2010;38:472–7.
- Adams RF, Gray W, Davies RJ, *et al*. Percutaneous image-guided cutting needle biopsy of the pleura in the diagnosis of malignant mesothelioma. *Chest* 2001;120:1798–802.
- McLean AN, Bicknell SR, McAlpine LG, *et al*. Investigation of pleural effusion: an evaluation of the new Olympus LTF semiflexible thoracofiberscope and comparison with Abram's needle biopsy. *Chest* 1998;114:150–3.
- Schönfeld N, Schwarz C, Kollmeier J, *et al*. Narrow band imaging (NBI) during medical thoracoscopy: first impressions. *J Occup Med Toxicol* 2009;4:24.
- Pathak V, Shepherd RW, Hussein E, *et al*. Safety and Feasibility of Pleural Cryobiopsy Compared to Forceps Biopsy During Semi-rigid Pleuroscopy. *Lung* 2017;195:371–5.
- Wang X-B, Yin Y, Miao Y, *et al*. Flex-rigid pleuroscopic biopsy with the SB knife Jr is a novel technique for diagnosis of malignant or benign fibrothorax. *J Thorac Dis* 2016;8:E1555–9.
- Shafiq M, Sethi J, Ali MS, *et al*. Pleural Cryobiopsy: A Systematic Review and Meta-Analysis. *Chest* 2020;157:223–30.
- Wang F, Wang Z, Tong Z, *et al*. A pilot study of autofluorescence in the diagnosis of pleural disease. *Chest* 2015;147:1395–400.
- Wang F, Zhou L, Wang Z, *et al*. Autofluorescence imaging-assisted medical thoracoscopy in the diagnosis of malignant pleural disease. *Respir Med* 2023;207:107114.
- Bonhomme O, Heinen V, Detrembleur N, *et al*. Probe-based confocal laser endomicroscopy for pleural malignancies diagnosis. *Respirology* 2021;26:188–95.
- Wijmans L, Baas P, Sieburgh TE, *et al*. Confocal Laser Endomicroscopy as a Guidance Tool for Pleural Biopsies in Malignant Pleural Mesothelioma. *Chest* 2019;156:754–63.
- Wurps H, Schönfeld N, Bauer TT, *et al*. Intra-patient comparison of parietal pleural biopsies by rigid forceps, flexible forceps and cryoprobe obtained during medical thoracoscopy: a prospective series of 80 cases with pleural effusion. *BMC Pulm Med* 2016;16:98.
- Light RW, Rogers JT, Cheng D, *et al*. Large Pleural Effusions Occurring after Coronary Artery Bypass Grafting. *Ann Intern Med* 1999;130:891–6.
- Koegelenberg CFN, Iruen EM, von Groote-Bidlingmaier F, *et al*. The utility of ultrasound-guided thoracentesis and pleural biopsy in undiagnosed pleural exudates. *Thorax* 2015;70:995–7.
- Zhang Y, Tang J, Zhou X, *et al*. Ultrasound-guided pleural cutting needle biopsy: accuracy and factors influencing diagnostic yield. *J Thorac Dis* 2018;10:3244–52.
- Bonhomme O, Duysinx B, Heinen V, *et al*. First report of probe based confocal laser endomicroscopy during medical thoracoscopy. *Respir Med* 2019;147:72–5.
- Gonzalez AV, Silvestri GA, Korevaar DA, *et al*. Assessment of Advanced Diagnostic Bronchoscopy Outcomes for Peripheral Lung Lesions: A Delphi Consensus Definition of Diagnostic Yield and Recommendations for Patient-centered Study Designs. An Official American Thoracic Society/American College of Chest Physicians Research Statement. *Am J Respir Crit Care Med* 2024;209:634–46.
- Ost DE, Feller-Kopman DJ, Gonzalez AV, *et al*. Reporting Standards for Diagnostic Testing: Guidance for Authors From Editors of Respiratory, Sleep, and Critical Care Journals. *J Bronchology Interv Pulmonol* 2023;30:207–22.
- Chen C-H, Cheng W-C, Wu B-R, *et al*. Feasibility and Safety of Pleuroscopic Cryobiopsy of the Pleura: A Prospective Study. *Can Respir J* 2018;2018:6746470.
- Sundaralingam A, Aujayeb A, Akca B, *et al*. Achieving Molecular Profiling in Pleural Biopsies: A Multicenter, Retrospective Cohort Study. *Chest* 2023;163:1328–39.
- Tsim S, Paterson S, Cartwright D, *et al*. Baseline predictors of negative and incomplete pleural cytology in patients with suspected pleural malignancy - Data supporting "Direct to LAT" in selected groups. *Lung Cancer (Auckl)* 2019;133:123–9.
- Schwarz C, Lübbert H, Rahn W, *et al*. Medical thoracoscopy: hormone receptor content in pleural metastases due to breast cancer. *Eur Respir J* 2004;24:728–30.
- Rozman A, Camlek L, Marc Malovrh M, *et al*. Feasibility and safety of parietal pleural cryobiopsy during semi-rigid thoracoscopy. *Clin Respir J* 2016;10:574–8.
- Liu D, Lu Y, Hu Z, *et al*. Malignant Pleural Effusion Supernatants Are Substitutes for Metastatic Pleural Tumor Tissues in EGFR Mutation Test in Patients with Advanced Lung Adenocarcinoma. *PLoS ONE* 2014;9:e89946.
- Giri M, Dai H, Guo S, *et al*. Efficacy and Safety of Pleural Cryobiopsy vs. Forceps Biopsy for Evaluation of Undiagnosed Pleural Effusion: A Systematic Review and Meta-Analysis. *Front Med (Lausanne)* 2022;9:847146.
- Masai K, Sasada S, Izumo T, *et al*. Pleuroscopic punch biopsy using insulated-tip diathermic knife-2 for the diagnosis of desmoplastic malignant mesothelioma. *J Bronchology Interv Pulmonol* 2013;20:345–8.
- Monici M. Cell and tissue autofluorescence research and diagnostic applications. *Biotechnol Annu Rev* 2005;11:227–56.

- 34 Chrysanthidis MG, Janssen JP. Autofluorescence videothoracoscopy in exudative pleural effusions: preliminary results. *Eur Respir J* 2005;26:989–92.
- 35 Porfyridis I, Georgiadis G, Michael M, et al. Rapid on-site evaluation with the Hemacolor rapid staining method of medical thoracoscopy biopsy specimens for the management of pleural disease. *Respirology* 2016;21:1106–12.
- 36 Chow SC, Shao J, Wang H. *Sample size calculations in clinical research*. 2nd edn. Boca Raton, Florida: Chapman & Hall/CRC, 2008.
- 37 Farrington CP, Manning G. Test statistics and sample size formulae for comparative binomial trials with null hypothesis of non-zero risk difference or non-unity relative risk. *Stat Med* 1990;9:1447–54.
- 38 Fleiss JL, Levin B, Paik MC. *Statistical methods for rates and proportions*. 3rd edn. New York: John Wiley & Sons, 2003.
- 39 Gart JJ, Nam J. Approximate interval estimation of the ratio of binomial parameters: a review and corrections for skewness. *Biometrics* 1988;44:323–38.
- 40 Gart JJ, Nam JM. Approximate interval estimation of the difference in binomial parameters: correction for skewness and extension to multiple tables. *Biometrics* 1990;46:637–43.
- 41 Julious SA, Campbell MJ. Tutorial in biostatistics: sample sizes for parallel group clinical trials with binary data. *Stat Med* 2012;31:2904–36.
- 42 Lachin JM. *Biostatistical methods*. New York: John Wiley & Sons, 2000.
- 43 Machin D, Campbell M, Fayers P, et al. *Sample size tables for clinical studies*. 2nd edn. Malden, Mass: Blackwell Science, 1997.
- 44 Miettinen OS, Nurminen M. Comparative analysis of two rates. *Stat Med* 1985;4:213–26.